

D321

AWS Developer

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A developer at a travel startup with 25 engineers puts an IAM user's access key and secret key in a Lambda function's source code so it can read from DynamoDB. The repository is private. The function runs 10,000 times a day. What is the recommended approach?

- A. Use the root account keys so permissions never fail.
- B. Keep the keys because the repository is private.
- C. Move the keys to environment variables in plain text.
- D. Remove the keys and grant DynamoDB read access through the function's execution role.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A fitness app with 2 million downloads lets users sign in with email or a social account. After sign-in, the mobile app must upload workout photos directly to S3. The app's color theme is orange. Which Cognito setup fits?

- A. An IAM user for each app user
- B. A user pool alone
- C. An identity pool alone with no sign-in
- D. A user pool for sign-in plus an identity pool to exchange tokens for temporary AWS credentials

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A web app only needs users to sign up, sign in, and receive JWTs that its own API Gateway endpoints check. The browser never calls AWS services directly. What is required?

- A. A Cognito identity pool
- B. Both, because an identity pool is always required
- C. A Cognito user pool
- D. An S3 bucket policy

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An orders table in DynamoDB holds 50 million items with partition key customerId and sort key orderDate. The table was created by a former teammate. A developer needs one customer's orders from the last 30 days. Which operation is most efficient?

- A. GetItem on customerId alone
- B. Scan with a filter on customerId and orderDate
- C. Scan, because filters make it as cheap as Query
- D. Query on customerId with a sort key condition on orderDate

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A team must find all items in a table where a non-key attribute, status, equals failed, and this query will run often. What makes the read efficient?

- A. Scan with a filter each time.
- B. Query the base table on status.
- C. Create a global secondary index with status as its partition key and Query it.
- D. Increase read capacity so Scan is fast.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A team deploys a new Lambda version with CodeDeploy using the setting Canary10Percent5Minutes. The team lead just returned from a conference. How is traffic shifted?

- A. 10% shifts to the new version, and after 5 minutes the remaining 90% shifts.
- B. All traffic shifts at once to a new environment.
- C. 10% more traffic shifts every 5 minutes until 100%.
- D. 5% shifts, and after 10 minutes the rest shifts.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A team wants a deployment that keeps the old environment fully running so it can switch all traffic back instantly if the new one fails. Which strategy fits?

- A.** All-at-once
- B.** Blue/green
- C.** Canary
- D.** Rolling

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A payments team using CodePipeline, which was set up during a company hackathon, currently runs unit tests after the deploy stage pushes to production. The team has 300 unit tests that run in 90 seconds. Where should the unit tests run?

- A.** Nowhere, because 300 tests are too slow.
- B.** In the build or test stage, before any deploy stage, so failures stop the pipeline.
- C.** Only in the source stage by developers, with no pipeline test.
- D.** After production deploy, as now.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A healthcare app written in Node.js must encrypt 25 MB patient PDF files with a KMS key before storing them in S3. The security architect holds three certifications. How should the app encrypt the files?

- A.** Split each file into 25 pieces and call Encrypt on each.
- B.** Call GenerateDataKey, encrypt the file locally with the plaintext data key, and store the encrypted data key with the file.
- C.** Store the plaintext data key next to the file.
- D.** Send each 25 MB file to the KMS Encrypt API.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

An app must read a large file that was protected with envelope encryption. The encrypted data key is stored with the file. What does the app do first?

- A.** Call KMS Decrypt on the encrypted data key, then decrypt the file locally with the returned plaintext key.
- B.** Export the KMS key and decrypt the file with it.
- C.** Call GenerateDataKey to get the key back.
- D.** Send the whole encrypted file to KMS Decrypt.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A checkout request at an online store that sells 5,000 items passes through API Gateway, three Lambda functions, and DynamoDB. Customers report checkouts taking 8 seconds. Each service's logs show no errors. What best identifies which step is slow?

- A.** Nothing is wrong, because the logs show no errors.
- B.** Raise every Lambda's memory to the maximum.
- C.** Enable AWS X-Ray tracing and inspect the service map and segment timings.
- D.** Read each Lambda's CloudWatch Logs one by one.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A data loader built by a two-person team writes to DynamoDB and occasionally receives ProvisionedThroughputExceededException during nightly imports of 2 million rows. How should it retry?

- A.** Retry with exponential backoff and jitter, up to a maximum.
- B.** Retry after a fixed 2-second wait, with every client using the same wait.
- C.** Retry immediately in a loop until it succeeds.
- D.** Never retry; throttled writes are always lost.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A developer at a travel startup ~~with 25 engineers~~ puts an IAM user's access key and secret key in a Lambda function's source code so it can read from DynamoDB. The repository is private. ~~The function runs 10,000 times a day.~~ What is the recommended approach?

Target word: how a Lambda function should get credentials **Decides:** Lambda gets temporary credentials from its execution role.

A	Inverted operation	The opposite of least privilege.
B	Camouflage	Private repos leak; keys in code remain a risk.
C	Dropped element	Still long-term keys; the role is the missing piece.
D	Correct	The SDK picks up role credentials automatically.

2. Answer D Concept 2

A fitness app ~~with 2 million downloads~~ lets users sign in with email or a social account. After sign-in, the mobile app must upload workout photos directly to S3. ~~The app's color theme is orange.~~ Which Cognito setup fits?

Target word: which Cognito component grants AWS access **Decides:** Direct S3 access from the client needs temporary AWS credentials.

A	Confusion set	IAM users do not scale to millions of app users.
B	Dropped element	Issues tokens but not AWS credentials for S3.
C	Adjacent category	Would give guest credentials without the required user sign-in.
D	Correct	The identity pool issues credentials scoped by an IAM role.

3. Answer C Concept 2 · transfer

A web app only needs users to sign up, sign in, and receive JWTs that its own API Gateway endpoints check. The browser never calls AWS services directly. What is required?

Target word: minimal Cognito component for sign-in only **Decides:** Tokens for the app's own API, no direct AWS calls.

A	Inverted operation	Issues AWS credentials, which are not needed here.
B	Absolute language	Identity pools are needed only for direct AWS access.
C	Correct	Handles sign-up and sign-in and issues JWTs.
D	Adjacent category	Controls bucket access, not user sign-in.

4. Answer D Concept 3

An orders table in DynamoDB holds 50 million items with partition key customerId and sort key orderDate. ~~The table was created by a former teammate.~~ A developer needs one customer's orders from the last 30 days. Which operation is most efficient?

Target word: efficient read pattern **Decides:** Partition key known and a sort key range.

A	Dropped element	GetItem needs the full primary key, including the sort key.
B	Confusion set	Reads all 50 million items, then filters.
C	Absolute language	Filters apply after capacity is consumed.
D	Correct	Reads only that customer's matching items.

5. Answer C Concept 3 · transfer

A team must find all items in a table where a non-key attribute, status, equals failed, and this query will run often. What makes the read efficient?

Target word: enable Query on a non-key attribute **Decides:** Query needs the attribute to be a key.

A	Inverted operation	Reads the whole table on every run.
B	Dropped element	Query requires a key attribute; status is not one.
C	Correct	A GSI makes status queryable.
D	Adjacent category	Pays more for the same wasteful pattern.

6. Answer A Concept 4

A team deploys a new Lambda version with CodeDeploy using the setting Canary10Percent5Minutes. ~~The team lead just returned from a conference.~~ How is traffic shifted?

Target word: interpret the canary setting **Decides:** Canary: a small share first, then the rest at once.

A	Correct	Canary moves traffic in two steps.
B	Adjacent category	That is all-at-once or blue/green style.
C	Confusion set	That describes a linear configuration.
D	Inverted operation	Swaps the percentage and the interval.

7. Answer B Concept 4 · transfer

A team wants a deployment that keeps the old environment fully running so it can switch all traffic back instantly if the new one fails. Which strategy fits?

Target word: reverse: strategy from the rollback need **Decides:** Two full environments with a traffic switch.

A	Inverted operation	Replaces everything with no standing old environment.
B	Correct	The old (blue) environment stays live for instant rollback.
C	Confusion set	Tests with a small share of traffic, but does not require keeping a full old environment.
D	Adjacent category	Old instances are replaced in batches; rollback means redeploying.

8. Answer B Concept 5

A payments team using CodePipeline, ~~which was set up during a company hackathon,~~ currently runs unit tests after the deploy stage pushes to production. ~~The team has 300 unit tests that run in 90 seconds.~~ Where should the unit tests run?

Target word: test placement in CI/CD **Decides:** Tests must gate deployment.

A	Camouflage	90 seconds is fast; test count is not the issue.
B	Correct	Fail fast before code reaches production.
C	Dropped element	Removes the automated gate.
D	Inverted operation	Bad code reaches customers before tests run.

9. Answer B Concept 6

A healthcare app ~~written in Node.js~~ must encrypt 25 MB patient PDF files with a KMS key before storing them in S3. ~~The security architect holds three certifications.~~ How should the app encrypt the files?

Target word: encrypting data larger than 4 KB with KMS **Decides:** KMS Encrypt handles only up to 4 KB.

A	Wrong stem number	1 MB pieces still exceed 4 KB.
B	Correct	Envelope encryption handles large data.
C	Inverted operation	Anyone with the file could decrypt it.
D	Confusion set	KMS Encrypt is limited to 4 KB of data.

10. Answer A Concept 6 · transfer

An app must read a large file that was protected with envelope encryption. The encrypted data key is stored with the file. What does the app do first?

Target word: reverse: decrypting an envelope-encrypted file **Decides:** The data key must be recovered before the file can be decrypted.

A	Correct	KMS unwraps the small data key; the bulk decryption happens locally.
B	Adjacent category	Treats the KMS key like a local key; KMS keys never leave KMS in plaintext.
C	Inverted operation	Creates a brand-new key that cannot decrypt the old file.
D	Confusion set	KMS handles only small payloads; the file was encrypted locally.

11. Answer C Concept 7

A checkout request at an online store ~~that sells 5,000 items~~ passes through API Gateway, three Lambda functions, and DynamoDB. Customers report checkouts taking 8 seconds. ~~Each service's logs show no errors.~~ What best identifies which step is slow?

Target word: tool for cross-service latency **Decides:** Latency across a chain of services.

A	Camouflage	Slowness can occur without errors.
B	Dropped element	Skips finding the root cause.
C	Correct	Traces show time spent in each hop.
D	Adjacent category	Shows component events, not end-to-end timing.

12. Answer A Concept 8

A data loader ~~built by a two-person team~~ writes to DynamoDB and occasionally receives ProvisionedThroughputExceededException during nightly imports ~~of 2 million rows~~. How should it retry?

Target word: retry pattern for throttling **Decides:** Throttling calls for spacing out retries.

A	Correct	Spreads retries so load drops below the limit.
B	Dropped element	Leaves out growth and jitter, so clients retry in sync.
C	Inverted operation	Adds load and prolongs throttling.
D	Absolute language	Throttling is transient and retryable.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Credentials in code	Let the SDK find credentials through the default provider chain (an IAM role on EC2, Lambda, or ECS); never hard-code access keys in source or config files.	Embedding access keys in the repository because it is private.	An app needs to call an AWS API and keys appear in the code.
2	Cognito user pools versus identity pools	User pools authenticate users and issue tokens (sign-up, sign-in); identity pools exchange tokens for temporary AWS credentials to call AWS services directly.	Using a user pool alone to give a mobile app direct S3 access.	Sign-in versus access to AWS resources from the client.
3	DynamoDB Query versus Scan	Query reads items by partition key (and optional sort key condition) efficiently; Scan reads the whole table and then filters, consuming capacity for every item read.	Using Scan with a filter as if it were cheap.	Fetching items for one known key in a large table.
4	Deployment strategies	All-at-once is fastest with downtime risk; rolling updates in batches; blue/green shifts traffic to a new environment for instant rollback; canary shifts a small percentage first, then the rest.	Calling a 10%-then-90% shift linear or blue/green.	Traffic percentages and timing in the deployment configuration.
5	CI/CD pipeline stages	Source, build, test, deploy; fast unit tests run early in the build stage so failures stop the pipeline before deployment.	Running tests after deploying to production.	Where a test step belongs in the pipeline.
6	KMS envelope encryption	KMS encrypts up to 4 KB directly; for larger data, generate a data key, encrypt the data locally with it, and store the encrypted data key alongside the data.	Sending a large file straight to KMS Encrypt.	A file larger than 4 KB to encrypt with KMS.
7	Tracing versus logging for root cause	CloudWatch Logs and metrics show what happened in each component; X-Ray traces a request across services to find where latency or errors start.	Reading one service's logs to find a slowdown that spans several services.	A request path through many services and an unknown slow step.
8	Retries with exponential backoff	On throttling or transient errors, retry with exponentially increasing waits plus random jitter, up to a limit.	Retrying immediately in a tight loop.	Throttling errors such as ProvisionedThroughputExceededException or HTTP 429.

Course scope (public description): AWS Developer examines the skills and knowledge needed to effectively implement automated and continuous testing processes for software deployments with cloud solutions. Students will learn to design software with Amazon Web Services (AWS), software development kits (SDKs), and command line interface (CLI), and to implement authentication, encryption, and authorization within an AWS environment. Students will also learn to design cloud service deployments with AWS infrastructure services, platform services,

and features. Students will learn skills to monitor automated testing for quality control and to perform root cause analysis on testing or production failures. There are no prerequisites for this course.